OOS Number: OOS- 262333

Phase I

Section A – Test Details					
Initiator:	Cuong Du (Written by Qiyue (	**Test Date:**	12-Oct-2026	**Date Initiated:**	12-Oct-2026
Date of Incident:	12-Oct-2026	**Test Name:**	Celsis Sterility Test	**Sample ID:**	ETX-261002-0288
Name of Analyst who Performed the Test:	Prepping Analyst: Mukyung Jang (MJ) Processing Analyst: Alex Saravia (ES) Aliquoting Analyst: Cuong Du (CCD) Reading Analyst: Cuong Du (CCD)	**Sample / Active Name**	TESAMORELIN 8 MG/ML (3 ML) Injection	**Lot #:**	LG342011076
		Dosage Form	Injectable		
Description of Incident:	On 12Oct26, sample ETX-261002-0288 was found positive for microbial growth after Celsis sterility testing.	☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method #: MICRO-SOP-44 (13) MICRO-SOP-53 (04)	Effective Date: 03-Jan-2025 24-Jul-2026	SOP / Test Method Rev: Rev: 13 Rev: 04	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 12-Oct-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑Yes ☐ No ☐ N/A	Yes, analysts Mukyung Jang, Alex Saravia, and Cuong Du were interviewed comprehensively.	
Correct samples analyzed?	☑Yes ☐ No ☐ N/A	Yes, Sample ID ETX-261002-0288	
Correct SOP / Test Method used/followed?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44	
Correct sample collection/technique?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑Yes ☐ No ☐ N/A	Yes, Information is available in EagleTrax under ETX-261002-0288	
Is the analyst qualified to perform the test?	☑Yes ☐ No ☐ N/A	Yes, all analysts are trained and qualified by quality to perform the test.	
Is formula worksheet provided and verified?	☐Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑Yes ☐ No ☐ N/A	Yes, the sample was stored refrigerated as per client's instructions	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐Yes ☑ No ☐N/A		
Was the sample transported appropriately?	☑Yes ☐ No ☐N/A		
Are the other test results from this run under investigation?	☐Yes ☑ No ☐N/A		

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Appropriate/correct glassware used?	☐Yes ☐ No ☑N/A		
Are reagents stored in proper condition and not expired?	☑Yes ☐ No ☐N/A	Materials and Reagents TSB (300mL): Lot: 08132026-1 FTM (300mL): Lot: 08182026-6 Tween (Polysorbate) 80: Lot: 0000743947 Celsis AmpiScreen Reagents: 022601-1400 Celsis Daily Wash & Rinse Kit: 012503-1142N Celsis ATP Positive Control: 022601-1483 TSA and Surface Plates: Please refer to the Environmental Logs uploaded under the submission on EagleTrax Monthly Cleaning: HaloMist: Lot: 2606A H2O2 strips: Lot: 260620	Materials and Reagents TSB: Exp: 11Nov2026 FTM: Exp: 16Nov2026 Tween 80: 31Jan2030 Celsis AmpiScreen Reagents: Jan 2027 Celsis Daily Wash & Rinse Kit: Jan 2027 Celsis ATP Positive Control: Jan 2027 TSA and Surface Plates: Please refer to the Environmental Logs uploaded under the submission on EagleTrax Monthly Cleaning: HaloMist: Exp: 06Jun28 H2O2 strips: Exp: 20Jun28
Standard Used	Celsis ATP Positive Control	022601-1483	Jan 2027
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐Yes ☐ No ☑N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐Yes ☐ No ☑N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑Yes ☐ No ☐ N/A	E002222	Dec 2026
Is the clean room facility certified?	☑Yes ☐ No ☐ N/A	E001736 (CR114), BSC E001316	Dec 2026, Jun 2027
System Performance Check (if applicable)?	☑Yes ☐ No ☐ N/A	E002222	Dec 2026
System Suitability RSD < 2% (PDA)?	☐Yes ☐ No ☑ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐Yes ☐ No ☑ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑Yes ☐ No ☐ N/A	Incubator E001034 (Monitored by Sensor E001501) Incubator E001031 (Monitored by Sensor E001505) ISO 5 BSC E001316 ISO 5 BSC E001798	Aug 2027 Feb 2027 Aug 2027 Feb 2027 Jun 2027 Jun 2027

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Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not Applicable	Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☑ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A QYC 22Oct26

Section D – Phase I Summary

Phase I Summary:

All analysts involved in the preparation, processing, aliquoting, and reading of the samples, specifically Mukyung Jang, Alex Saravia, and Cuong Du, were interviewed comprehensively. Their responses are documented throughout this report. This investigation was performed as per MICRO-SOP-53, Sterility Test Out-of-Specification Investigation Procedure.

Upon arrival on 02Oct26, the sample was stored refrigerated in accordance with the client instructions. Both sample prepping analyst Mukyung Jang and processing analyst Alex Saravia inspected the sample during preparation and processing. No leaks, cracks, or visible turbidity were observed, and container integrity remained intact prior to testing.

All reagents and supplies mentioned in the material section above were stored according to supplier recommendations, and their integrity was visually verified before utilization. Moreover, each reagent, media, and consumable possessed valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom suite used for processing procedures, Cleanroom Suite 114, comprises three interconnected sections. These include the innermost ISO 7 cleanroom 114B, which connects to the middle ISO 7 buffer room 114A, and extends to the outermost ISO 8 anteroom 114. Outward-cascading positive air pressure differentials are continuously maintained across the suite to ensure controlled, unidirectional airflow from cleanroom 114B through buffer room 114A and into anteroom 114.

The innermost ISO 7 cleanroom 114B and its ISO 5 BSC E001316 were thoroughly cleaned and prepared before initiating testing by analyst Alex Saravia as per MICRO-SOP-2 for Environmental Monitoring of the Clean-room Facility and MICRO-SOP-9 for Cleaning and Disinfecting Procedure for Microbiology. For Celsis aliquoting, intermediate ISO 7 buffer room 114A and its ISO 5 BSC E001798 were thoroughly cleaned and prepared by analyst Cuong Du before initiating testing as per MICRO-SOP-44 and MICRO-SOP-9. Both biosafety cabinets, E001316 in cleanroom 114B and E001798 in buffer room 114A, were certified and approved by the Engineering and Quality Assurance teams prior to use.

On 05Oct26, the sample vials for ETX-261002-0288 were retrieved from controlled refrigerated storage and transferred into the sterile microbiology laboratory suites. In accordance with standard operating procedures, container surfaces were sprayed with acidified bleach disinfectant, placed into pre-disinfected transfer bins, and allowed a verified 10-minute contact time. Secondary surface disinfection with acidified bleach was conducted in the ISO 8 anteroom 114 prior to entering cleanroom 114B. Inside ISO 7 cleanroom 114B, processing analyst Alex Saravia executed a final disinfection step before introducing the containers into the ISO 5 BSC E001316. Direct inoculation was performed by adding 15 mL of sample (divided across ten 3 mL vials) along with 3 mL of sterile Tween 80 into 300 mL of Fluid Thioglycollate Medium (FTM) and 300 mL of Soybean-Casein Digest Medium (TSB) in accordance with method suitability study ETX-250826-0390. A matching negative control jar with identical modifications was prepared concurrently and handled identically. Following inoculation, media bottles were transferred into calibrated incubators to commence incubation.

Upon completion of incubation on 12Oct26, the media bottles for ETX-261002-0288 were surface disinfected and transferred to the intermediate ISO 7 buffer room 114A for aliquoting as per MICRO-SOP-44, Celsis Sterility Testing. In buffer room 114A, containers underwent secondary disinfection before introduction into the ISO 5 BSC E001798. Within the BSC, analyst Cuong Du aseptically aliquoted samples into assay cuvettes. Subsequently, Celsis sterility readouts were performed by Cuong Du on instrument Celsis Advance 2 E002222 under workload 101226-2222-5.wkl.

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Phase I Summary (continued):

Following instrument analysis on 12Oct26, sample ETX-261002-0288 was found to yield positive luminescence readings in the 1 x 300 mL FTM media jar. The FTM sample yielded an average luminescence of 3,647,568 RLU (duplicate tube readings of 3,608,783 RLU and 3,686,352 RLU, CV% 1.5%), significantly exceeding the positive cut-off threshold of 2,596.5 RLU. In contrast, the 1 x 300 mL TSB media jar yielded negative readings well below its positive cut-off threshold of 2,326.5 RLU. All instrument system controls and background checks met defined acceptance criteria, including the instrument blank (29 RLU, cut-off 100.0 RLU), reagent blank (103 RLU, cut-off 1,000.0 RLU), and Celsis ATP Positive Control (99,973 RLU, cut-off 50,000.0 RLU, passing specification).

Following the OOS result, the positive FTM media jar for ETX-261002-0288 was submitted for Differential Staining and Microbial Identification under subculture submission ETX-261012-0654. Third-generation Oxford Nanopore sequencing confirmed the isolate as Bacillus paralicheniformis (Gram-positive rods, spore-forming, 85.73% match), as detailed in Table 1.

All culture media utilized were within their approved manufacturer expiration dates. Negative control media jars were prepared, manipulated, and incubated alongside the test sample under identical conditions, and all negative controls exhibited no microbial recovery (FTM negative control 866 RLU, TSB negative control 776 RLU).

Monthly cleaning and disinfection of Cleanroom Suite 114 (including ISO 8 anteroom 114, ISO 7 buffer room 114A, ISO 7 cleanroom 114B, and the processing BSCs E001316 and E001798) were performed on 27Sep26 by Sasha Allen (SAA), as per MICRO-SOP-9, Cleaning and Disinfecting Procedure for Microbiology. Decontamination was performed using HaloMist (Lot 2606A, Exp 06Jun28), and all chemical indicator strips (Lot 260620, Exp 20Jun28) passed, verifying that environmental sanitization remained validated and in control prior to testing.

Attached Tables 2 and 3 present the environmental monitoring results spanning the testing period. Environmental monitoring (EM) plates were incubated for no less than 48 hours at 30 to 35 °C and no less than an additional five days at 20 to 25 °C, as per MICRO-SOP-2, Environmental Monitoring of the Clean-room Facility. Table 2 summarizes environmental monitoring during Celsis sample processing, and Table 3 summarizes monitoring during Celsis aliquoting.

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Phase I Summary (continued):

A review of environmental monitoring results during the sample processing phase showed complete absence of microbial growth across daily personnel touch plates (0 CFU for Alex Saravia), ISO 5 BSC E001316 work surface contact plates (0 CFU), and settling plates (0 CFU) on the processing date of 05Oct26, the preceding sampling date of 01Oct26, and the subsequent sampling date of 06Oct26. Weekly active air and surface monitoring of Cleanroom Suite 114 conducted on 08Oct26 demonstrated zero microbial recovery (No Growth) across all suite cleanrooms.

A review of environmental monitoring results during the aliquoting phase showed complete absence of microbial growth across daily personnel monitoring plates (0 CFU for Cuong Du), ISO 5 BSC E001798 surface plates (0 CFU), and settling plates (0 CFU) on the aliquoting date of 12Oct26, the preceding date of 09Oct26, and the subsequent date of 13Oct26. Weekly surface monitoring of Cleanroom Suite 114 conducted on 14Oct26 demonstrated zero microbial recovery (No Growth) across all cleanroom work surfaces. Weekly active air sampling conducted on 14Oct26 demonstrated no microbial growth in the ISO 7 cleanrooms (114A and 114B); however, 6 CFUs under sample ETX-261021-0578 were recovered from the outermost ISO 8 anteroom 114. Differential staining confirmed the 6 CFUs as 3 Gram-positive cocci, 2 Gram-variable coccobacilli, and 1 budding yeast (Microbial ID: Pending).

Regarding the evaluation of environmental monitoring recoveries, all direct product manipulations were conducted strictly within validated ISO 5 primary containment (BSC E001316 during processing and BSC E001798 during aliquoting). The 6 CFUs recovered during weekly active air sampling on 14Oct26 were strictly confined to the outermost ISO 8 anteroom 114, which functions as an outer personnel gowning and transition vestibule. There is minimal exposure of the test sample to ambient ISO 8 air, as samples and supplies are transferred in disinfected, closed containers on carts through the layered cleanroom suites. Outward-cascading positive air pressure differentials and physical barrier doors continuously prevent migration from ISO 8 into ISO 7 and ISO 5 spaces.

Furthermore, all 6 CFUs recovered from the ISO 8 anteroom (ETX-261021-0578) were confirmed by differential staining as Gram-positive cocci (3 CFUs), Gram-variable coccobacilli (2 CFUs), and budding yeast (1 CFU), all of which exhibit cellular morphologies different from the Gram-positive spore-forming rods (Bacillus paralicheniformis) recovered from the positive product sample. Additionally, complete absence of microbial recovery (No Growth) was confirmed across all critical ISO 7 cleanrooms (114A and 114B), processing surfaces, BSC settling plates, and analyst fingertip touch plates during both the processing and aliquoting sessions. This demonstrates that robust aseptic boundaries were maintained throughout the testing timeframe and confirms that no viable contamination transfer pathway existed between the exterior support areas and the critical ISO 5 processing environment.

All analysts demonstrated full adherence to aseptic protocols and cleaning procedures as outlined in MICRO-SOP-9 and MICRO-SOP-44. A comprehensive audit of equipment calibrations, incubator operational temperatures, and differential pressure logs confirmed that all facility and operational parameters remained within established validated specifications throughout the testing timeframe.

To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed on the same day. All other samples processed by the analyst that day tested negative. These findings suggest that cross-contamination between samples is highly unlikely.

Analyzing a 6-month sample history for Optimal Balance Pharmacy (E19193), this specific analyte "TESAMORELIN 8 MG/ML (3 ML) Injection" has had no prior Out-of-Specification (OOS) occurrences under Celsis sterility testing.

Therefore, the likelihood of laboratory error contributing to this failure is considered minimal, and the original result is deemed valid.

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Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes

☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Not Applicable

Prepared by (*Print*)**:**	Signature/Date:
Qiyue Chen	SIGN
Lab Manager/Designee (*Print*)**:**	Signature/Date:
Robin Seymour	SIGN

Section E – Quality Disposition

Can investigation be closed?

☐ Yes

☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date:
	SIGN

Table 1: Information for ETX-261002-0288 under investigation

Processing Analyst	Aliquoting Analyst	Sample ID	Related Microbial ID	Media with microbial growth	Microbial ID
Alex Saravia	Cuong Du	ETX-261002-0288	ETX-261012-0654	1 x 300mL FTM	*Bacillus paralicheniformis*

Table 2: Environmental Monitoring from Processing Performed on 05Oct26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst	Day /Week(s)	Observation	EM Plate ETX Number	Microbial ID	Notes
Personnel EM								
Personnel (Left Touch and Right Touch)	Daily	29Sep26	ES	Date Before Testing	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	05Oct26	ES	Date of Testing	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	06Oct26	ES	Date After Testing	No Growth	Not Applicable	None	None
Biological Safety Cabinet EM for E001316								
Surface Sampling of ISO 5 (4 locations)	Daily	01Oct26	QYC	Date Before Testing	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	05Oct26	ES	Date of Testing	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	06Oct26	ES	Date After Testing	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	01Oct26	QYC	Date Before Testing	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	05Oct26	ES	Date of Testing	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	06Oct26	ES	Date After Testing	No Growth	Not Applicable	None	None
Weekly Active Air Sampling of Cleanroom 114, 114A, 114B (Suite 114) with Processing BSCs								
Active Air Sampling of Cleanrooms	Weekly	08Oct26	SMO	Week of Testing	No Growth	Not Applicable	None	None
Weekly Surface Sampling of Cleanroom 114, 114A, 114B (Suite 114) with Processing BSCs								
Surface Sampling of Cleanrooms	Weekly	08Oct26	SMO	Week of Testing	No Growth	Not Applicable	None	None

Table 3: Environmental Monitoring from Aliquoting Performed on 12Oct26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst	Day /Week(s)	Observation	EM Plate ETX Number	Microbial ID	Notes
Personnel EM								
Personnel (Left Touch and Right Touch)	Daily	09Oct26	CCD	Date Before Aliquoting	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	12Oct26	CCD	Date of Aliquoting	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	13Oct26	CCD	Date After Aliquoting	No Growth	Not Applicable	None	None
Biological Safety Cabinet EM for E001798								
Surface Sampling of ISO 5 (4 locations)	Daily	09Oct26	CCD	Date Before Aliquoting	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	12Oct26	CCD	Date of Aliquoting	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	13Oct26	CCD	Date After Aliquoting	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	09Oct26	CCD	Date Before Aliquoting	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	12Oct26	CCD	Date of Aliquoting	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	13Oct26	CCD	Date After Aliquoting	No Growth	Not Applicable	None	None
Weekly Active Air Sampling of Cleanroom 114, 114A, 114B (Suite 114) with Processing BSCs								
Active Air Sampling of Cleanrooms	Weekly	14Oct26	SMO	Week of Testing	6 CFUs (ISO 8 114)	ETX-261021-0578	Gram (+) cocci Gram-variable coccobacilli Budding yeast	None
Weekly Surface Sampling of Cleanroom 114, 114A, 114B (Suite 114) with Processing BSCs								
Surface Sampling of Cleanrooms	Weekly	14Oct26	SMO	Week of Testing	No Growth	Not Applicable	None	None